

Exploring the Structure of Policies Produced by Markov Decision Processes for Redundant Systems

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1. Motivation

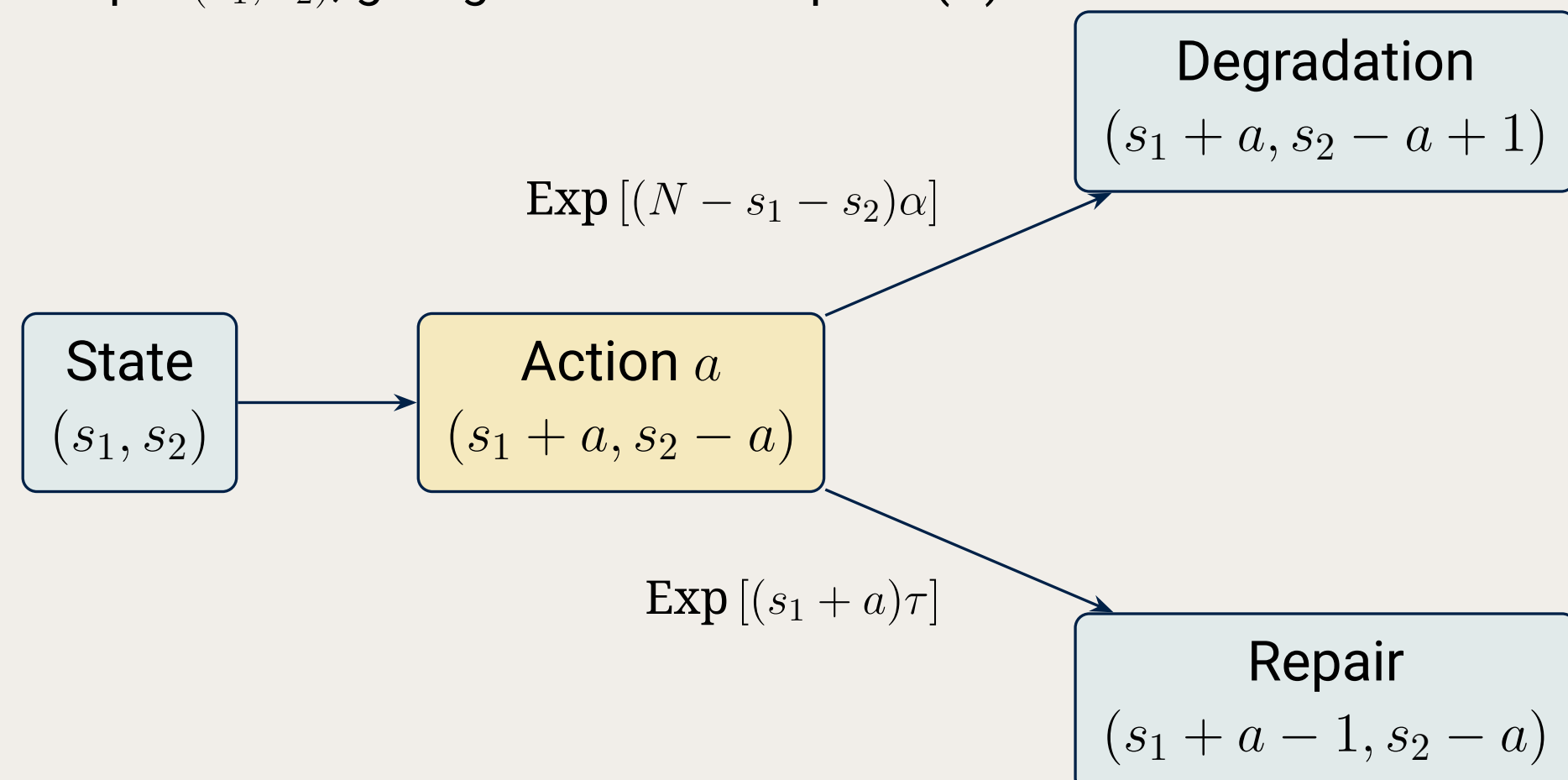
- In critical systems, we sometimes introduce redundancy, meaning that we install spare copies of components so that the system can survive component failures
- This leads to a question for system maintainers: when components fail, should we repair them, and if so, how many?
- Repairs are not free, so “always repair everything” is not always optimal

2. The Model

Each component has one of three statuses:

0. Healthy
1. Broken but being repaired
2. Broken and not being repaired

The number of components with each status can be described by a 2-tuple (s_1, s_2) , giving us our state space (S) .



- At each state we pick an action $a : 0 \leq a \leq s_2$, representing the number of components we wish to send to repair
- Our policy is a map from the state space S to the action space $\mathcal{A}$
- The system is online whenever the number of healthy components is greater than or equal to the minimum k
- We have a reward function that introduces a cost for each ongoing repair and a (large) penalty for system downtime

$$r(s, a) = -(s_1 + a)r - p\mathbb{1}_{\{s_0 < k\}}$$

- To find the optimal policy, we apply standard algorithms to maximise the long-run average reward of the model

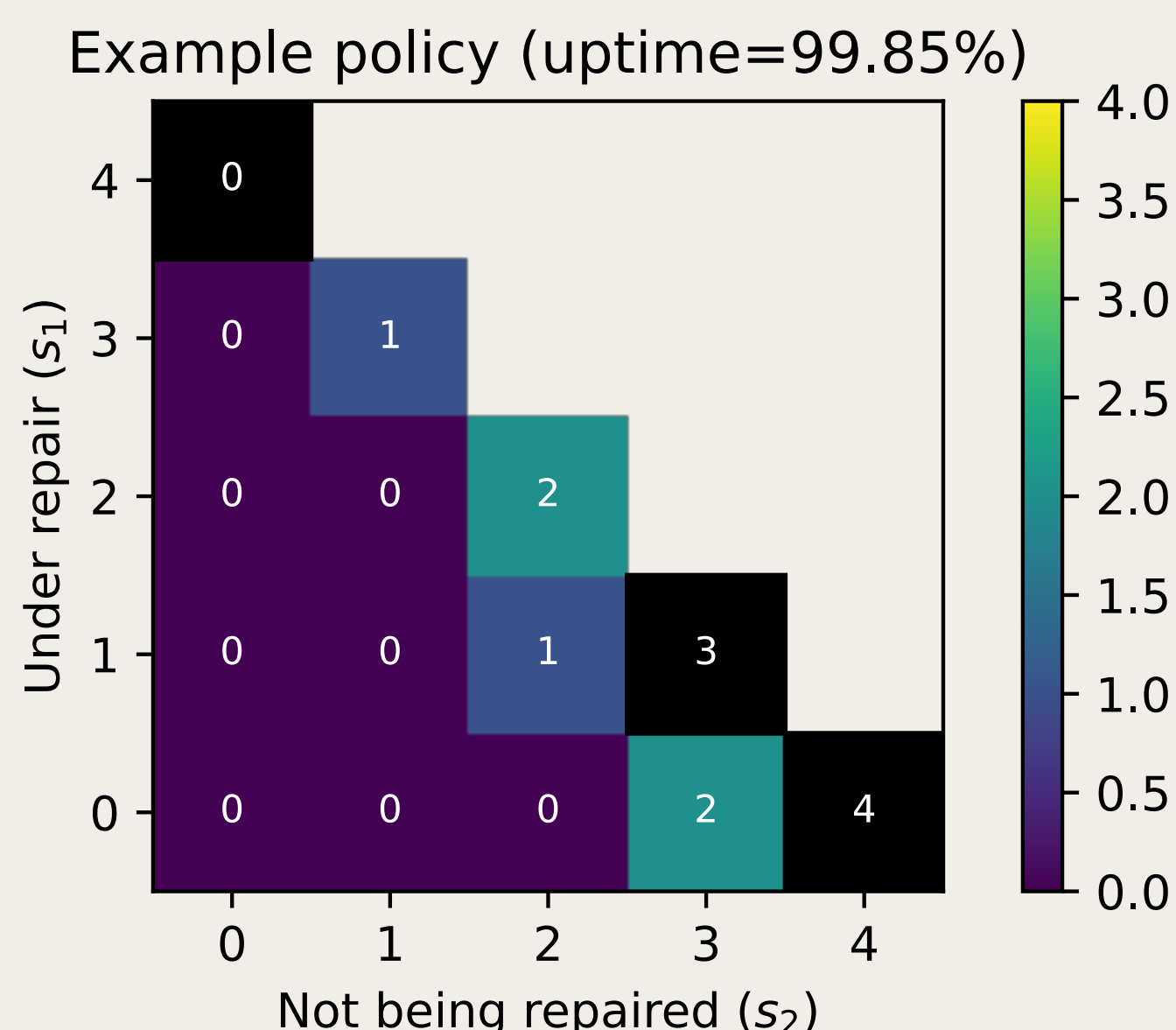
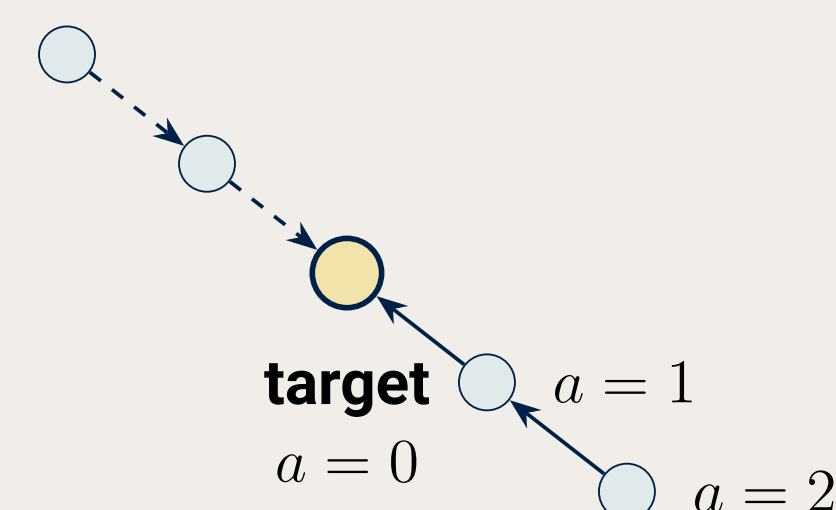


Figure: States shown in black are transient, so they do not contribute to the long-term average

This model can be extended to include different types of components.

3. Theoretical Results

Action targets Along a diagonal of the recurrent class, actions aim towards a common target state. In the extended model, where repairs may also be cancelled, this holds on both sides of the target.



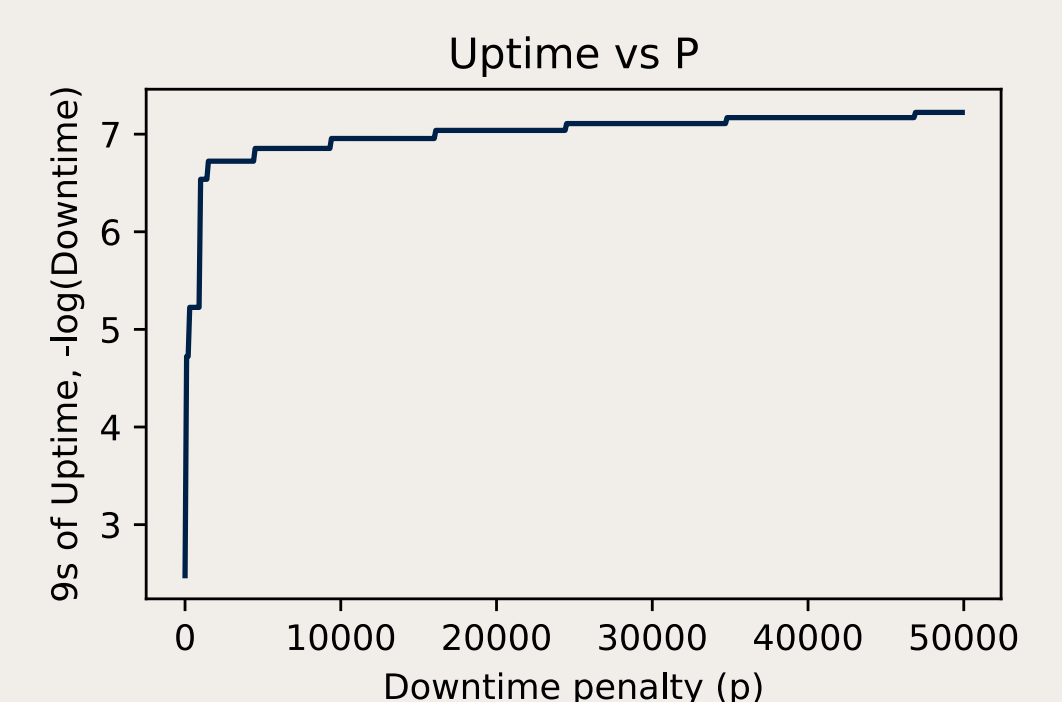
If $s = (s_1, s_2) \in R$, $0 \leq x \leq \pi(s)$, and $(s_1 + x, s_2 - x) \in R$ then

$$\pi(s_1 + x, s_2 - x) = \pi(s) - x.$$

Allowing cancellation extends this relation across the whole diagonal, reducing the description of an $O(N^2)$ policy to $O(N)$.

Uptime monotonicity

Increasing the penalty p for system downtime cannot decrease the uptime achieved by an optimal policy. If $p_1 < p_2$, π_1, π_2 the corresponding optimal policies, then $\text{Uptime}(\pi_1) \leq \text{Uptime}(\pi_2)$



Damage structure of the recurrent class Every recurrent class has a unique least-damaged state of the form $(0, \beta)$. Moreover, no intermediate damage level is skipped:

$$\{s_1 + s_2 : s \in R\} = \{\beta, \beta + 1, \dots, N\}.$$

Thus every recurrent class contains a state in which all N components are damaged, so 100% uptime is impossible.

4. Conjectures

All conjectures have been numerically tested across 11,760 problems
Zeros after action target An extension of the original action target result: if $x > a$, $(s_1 + x, s_2 - x) \in R$ then $\pi(s_1 + x, s_2 - x) = 0$. Proving this would allow us to store policies in $O(N)$ space.

Action monotonicity An extension of the uptime monotonicity result: Let $p_1 < p_2$ and π_1, π_2 be the corresponding optimal policies then $\pi_1(s) \leq \pi_2(s)$ for all states in the intersection of the recurrent classes. Showing this would give a convergence result (as $p \rightarrow \infty$), since actions are bounded.

Gravity of the recurrent class If $s = (s_1, s_2) \in R$ and $\forall s'_1 : 0 \leq s'_1 \leq s_1$ then $(s'_1, s_2) \in R$. Showing this would mean that we could describe the shape of the recurrent class in $O(N)$ space, as we would only need to give the state with the highest s_1 value in each column. We have currently proved this only for states with action 0.

5. Model Limitations

Independence We should not expect failures to be independent. Often, one event can stress many components simultaneously.

Homogeneity We assume that no component is preferable to any other. This is frequently not the case, for example data transfer speeds in a two-component-type model with HDDs and SSDs.

Switching We assume that switching to a healthy spare is instantaneous, does not stress the replacement, and never fails.

Distribution Decay and repair are modelled as exponential random variables, meaning that both are memoryless.